

DSA Logic Building – Master Roadmap

A gradual problem-solving journey from absolute beginner logic to strong DSA foundations. Solve the questions in order and focus on understanding before optimization.

STAGE 0: Variables and Simple Operations

1. Store your age and print it.
2. Store two numbers and print their sum.
3. Print subtraction of two numbers.
4. Print multiplication.
5. Print division.
6. Print remainder.
7. Swap two variables.
8. Store three numbers and print their sum.
9. Find average of two numbers.
10. Find average of three numbers.
11. Calculate square of a number.
12. Calculate cube of a number.
13. Calculate square root.
14. Calculate perimeter of a rectangle.
15. Calculate area of a rectangle.
16. Calculate area of a square.
17. Calculate area of a circle.
18. Convert Celsius to Fahrenheit.
19. Convert Fahrenheit to Celsius.
20. Convert kilometers to meters.
21. Convert meters to centimeters.
22. Convert hours to minutes.
23. Convert minutes to seconds.
24. Convert days to hours.
25. Calculate simple interest.
26. Calculate compound interest.
27. Calculate percentage.
28. Calculate discount.
29. Calculate final price after discount.
30. Calculate GST amount.
31. Calculate total bill with GST.

32. Calculate profit.
33. Calculate loss.
34. Calculate profit percentage.
35. Calculate salary after bonus.
36. Calculate salary after deduction.
37. Convert bytes to kilobytes.
38. Convert kilobytes to megabytes.
39. Convert MB to GB.
40. Find quotient and remainder.

STAGE 1: Comparison Thinking

41. Check whether a number is positive.
42. Check whether a number is negative.
43. Check whether a number is zero.
44. Check even or odd.
45. Find greater between two numbers.
46. Find smaller between two numbers.
47. Check whether two numbers are equal.
48. Find largest among three numbers.
49. Find smallest among three numbers.
50. Find middle number among three numbers.
51. Check whether a number is divisible by 2.
52. Check divisibility by 3.
53. Check divisibility by 5.
54. Check divisibility by 10.
55. Check divisibility by both 3 and 5.
56. Check divisibility by either 3 or 5.
57. Check whether a year is leap year.
58. Check voting eligibility.
59. Check driving eligibility.
60. Check whether marks are passing.
61. Assign grade based on marks.
62. Check whether a character is uppercase.
63. Check whether lowercase.
64. Check whether alphabet.
65. Check whether digit.
66. Check whether special character.

67. Check vowel or consonant.
68. Check whether a number lies between two numbers.
69. Check whether three sides form a triangle.
70. Identify triangle type.
71. Check whether a person is eligible for discount.
72. Find absolute value.
73. Find sign of multiplication of two numbers.
74. Find maximum using ternary operator.
75. Check whether number is multiple of another.
76. Check whether a number is a two-digit number.
77. Check whether a number is a three-digit number.
78. Check whether temperature is hot, cold, or normal.
79. Calculate electricity bill based on units.
80. Calculate income tax based on slabs.

STAGE 2: Loop Thinking

81. Print 1 to N.
82. Print N to 1.
83. Print 1 to 100.
84. Print even numbers.
85. Print odd numbers.
86. Print multiples of 3.
87. Print multiples of 5.
88. Print multiples of both 3 and 5.
89. Print numbers divisible by 7.
90. Print first N natural numbers.
91. Find sum from 1 to N.
92. Find sum from N to 1.
93. Find sum of even numbers.
94. Find sum of odd numbers.
95. Count numbers from 1 to N.
96. Find product from 1 to N.
97. Find factorial.
98. Find power.
99. Print multiplication table.
100. Print tables from 1 to 10.
101. Print squares from 1 to N.

- 102. Print cubes from 1 to N.
- 103. Sum of squares.
- 104. Sum of cubes.
- 105. Print numbers divisible by 2 but not 3.
- 106. Print numbers divisible by 3 but not 2.
- 107. Count even numbers from 1 to N.
- 108. Count odd numbers from 1 to N.
- 109. Count multiples of 7.
- 110. Sum numbers divisible by 5.
- 111. Product of even numbers from 1 to N.
- 112. Find largest among N inputs.
- 113. Find smallest among N inputs.
- 114. Find sum of N user inputs.
- 115. Find average of N inputs.
- 116. Count positive inputs.
- 117. Count negative inputs.
- 118. Count zeros.
- 119. Find largest positive number.
- 120. Find smallest negative number.

STAGE 3: While Loop and Digit Logic

- 121. Print numbers until user enters 0.
- 122. Keep taking numbers and find sum.
- 123. Keep taking numbers and find average.
- 124. Count positive numbers until 0.
- 125. Count negative numbers until 0.
- 126. Find largest number until 0.
- 127. Find smallest number until 0.
- 128. Keep asking for password until correct.
- 129. Create a simple login retry system.
- 130. Print numbers until N.
- 131. Find factorial using while loop.
- 132. Reverse a number using while.
- 133. Count digits using while.
- 134. Find sum of digits.
- 135. Find product of digits.
- 136. Find largest digit.

- 137. Find smallest digit.
- 138. Count even digits.
- 139. Count odd digits.
- 140. Sum even digits.
- 141. Sum odd digits.
- 142. Find first digit.
- 143. Find last digit.
- 144. Remove last digit repeatedly.
- 145. Print every digit separately.
- 146. Print digits in reverse order.
- 147. Check if all digits are even.
- 148. Check if all digits are odd.
- 149. Check if a number contains digit 0.
- 150. Count occurrences of a particular digit.

STAGE 4: Number Logic Laboratory

- 151. Reverse a number.
- 152. Check palindrome number.
- 153. Check Armstrong number.
- 154. Check strong number.
- 155. Check perfect number.
- 156. Check prime number.
- 157. Print all primes from 1 to N.
- 158. Count primes from 1 to N.
- 159. Find sum of primes from 1 to N.
- 160. Find largest prime below N.
- 161. Find smallest prime greater than N.
- 162. Find factorial.
- 163. Find GCD.
- 164. Find LCM.
- 165. Find GCD using Euclidean algorithm.
- 166. Check coprime numbers.
- 167. Generate Fibonacci series.
- 168. Find Nth Fibonacci.
- 169. Find sum of Fibonacci numbers.
- 170. Check if number belongs to Fibonacci.
- 171. Find sum of factorial digits.

172. Check Harshad number.
173. Check automorphic number.
174. Check neon number.
175. Check spy number.
176. Check sunny number.
177. Check duck number.
178. Check abundant number.
179. Check deficient number.
180. Find all perfect numbers in a range.
181. Reverse only the first half of digits.
182. Reverse only the last half.
183. Find second largest digit.
184. Find second smallest digit.
185. Find difference between largest and smallest digit.
186. Sort three digits manually.
187. Find frequency of every digit.
188. Remove all zeros from a number.
189. Replace digit 0 with 1.
190. Replace all even digits with 0.
191. Replace all odd digits with 1.
192. Create a number from reversed digits.
193. Check whether first and last digit are equal.
194. Find sum of first and last digit.
195. Find product of first and last digit.
196. Check whether all digits are unique.
197. Find duplicate digits.
198. Find most frequent digit.
199. Find least frequent digit.
200. Check whether digits are in increasing order.

STAGE 5: Nested Loop Thinking

201. Print a 3 x 3 grid.
202. Print coordinates of a matrix.
203. Print all pairs from 1 to N.
204. Print pairs where sum is 10.
205. Print pairs where product is even.
206. Print all combinations of two numbers.

- 207. Print all combinations of three numbers.
- 208. Count all pairs.
- 209. Count pairs with equal sum.
- 210. Find pairs divisible by 5.
- 211. Print multiplication table using nested loops.
- 212. Print a rectangle.
- 213. Print a square.
- 214. Print rows and columns.
- 215. Print matrix indices.
- 216. Print diagonal coordinates.
- 217. Print upper triangle coordinates.
- 218. Print lower triangle coordinates.
- 219. Count iterations in nested loops.
- 220. Create a simple multiplication matrix.

STAGE 6: Patterns

- 221. Print one star.
- 222. Print N stars in one line.
- 223. Print N rows of stars.
- 224. Print N columns.
- 225. Print square.
- 226. Print rectangle.
- 227. Print right triangle.
- 228. Print inverted triangle.
- 229. Print number triangle.
- 230. Print repeated number triangle.
- 231. Print alphabet triangle.
- 232. Print repeated alphabet triangle.
- 233. Print reverse number triangle.
- 234. Print odd number triangle.
- 235. Print even number triangle.
- 236. Print continuous numbers.
- 237. Print continuous alphabets.
- 238. Print multiplication pattern.
- 239. Print binary pattern.
- 240. Print alternating 0 and 1 pattern.
- 241. Print pyramid.

- 242. Print inverted pyramid.
- 243. Print full number pyramid.
- 244. Print reverse number pyramid.
- 245. Print alphabet pyramid.
- 246. Print hollow square.
- 247. Print hollow rectangle.
- 248. Print hollow triangle.
- 249. Print hollow inverted triangle.
- 250. Print hollow pyramid.
- 251. Print rhombus.
- 252. Print hollow rhombus.
- 253. Print diamond.
- 254. Print half diamond.
- 255. Print butterfly.
- 256. Print X pattern.
- 257. Print plus pattern.
- 258. Print hourglass.
- 259. Print sandglass.
- 260. Print Pascal's triangle.
- 261. Print number diamond.
- 262. Print alphabet diamond.
- 263. Print palindrome number pyramid.
- 264. Print symmetric number pyramid.
- 265. Print concentric square.
- 266. Print spiral matrix.
- 267. Print zigzag pattern.
- 268. Print snake pattern.
- 269. Print wave pattern.
- 270. Print checkerboard.

STAGE 7: Functions and Problem Decomposition

- 271. Create a function to add two numbers.
- 272. Create a function to subtract.
- 273. Create a function to multiply.
- 274. Create a function to divide.
- 275. Create a function to check even or odd.
- 276. Create a function to find maximum.

277. Create a function to find minimum.
278. Create a function to check prime.
279. Create a function to calculate factorial.
280. Create a function to reverse number.
281. Create a function to check palindrome.
282. Create a function for GCD.
283. Create a function for LCM.
284. Create a function for Fibonacci.
285. Create a function to calculate power.
286. Create a function to count digits.
287. Create a function to sum digits.
288. Create a function to find maximum digit.
289. Create a function to check Armstrong.
290. Create a function that returns boolean.
291. Build a calculator.
292. Build a number analyzer.
293. Build a student grade system.
294. Build a simple ATM.
295. Build a menu-driven calculator.
296. Build a prime checker menu.
297. Build a number statistics program.
298. Build a temperature converter.
299. Build a unit converter.
300. Build a mini utility application.

STAGE 8: Arrays From Absolute Zero

301. Create an array.
302. Print all elements.
303. Print first element.
304. Print last element.
305. Find array length.
306. Find sum.
307. Find average.
308. Find maximum.
309. Find minimum.
310. Count even elements.
311. Count odd elements.

- 312. Count positive elements.
- 313. Count negative elements.
- 314. Count zeros.
- 315. Find sum of even elements.
- 316. Find sum of odd elements.
- 317. Print array in reverse.
- 318. Copy an array.
- 319. Search an element.
- 320. Count occurrences of an element.
- 321. Find first occurrence.
- 322. Find last occurrence.
- 323. Check whether an element exists.
- 324. Replace an element.
- 325. Replace all occurrences.
- 326. Increment every element.
- 327. Double every element.
- 328. Find difference between consecutive elements.
- 329. Find sum of consecutive pairs.
- 330. Find largest difference between consecutive elements.
- 331. Check whether array is sorted ascending.
- 332. Check descending.
- 333. Reverse an array.
- 334. Swap first and last.
- 335. Swap adjacent elements.
- 336. Find second largest.
- 337. Find second smallest.
- 338. Find third largest.
- 339. Find frequency of every element.
- 340. Find duplicate elements.

STAGE 9: Array Logic Training

- 341. Remove duplicates.
- 342. Find unique elements.
- 343. Find first duplicate.
- 344. Find first non-repeating element.
- 345. Find most frequent element.
- 346. Find least frequent element.

- 347. Move zeros to end.
- 348. Move zeros to beginning.
- 349. Move negative numbers to beginning.
- 350. Move positive numbers to beginning.
- 351. Separate even and odd numbers.
- 352. Separate positive and negative numbers.
- 353. Reverse only even elements.
- 354. Reverse only odd elements.
- 355. Replace negative numbers with zero.
- 356. Find missing number.
- 357. Find multiple missing numbers.
- 358. Find duplicate in range 1 to N.
- 359. Find all duplicates.
- 360. Check whether two arrays are equal.

STAGE 10: Pair and Combination Thinking

- 361. Print all pairs.
- 362. Count all pairs.
- 363. Find pair with target sum.
- 364. Find all pairs with target sum.
- 365. Count pairs with target sum.
- 366. Find pair with target difference.
- 367. Find pair with target product.
- 368. Find pair with maximum sum.
- 369. Find pair with minimum difference.
- 370. Find closest pair to target.
- 371. Find three numbers with target sum.
- 372. Count triplets with target sum.
- 373. Find quadruplets with target sum.
- 374. Find pairs with equal values.
- 375. Find pairs where first is greater than second.
- 376. Find inversions using brute force.
- 377. Find maximum product pair.
- 378. Find minimum product pair.
- 379. Find pair with largest difference.
- 380. Find all subarrays.

STAGE 11: Subarray Thinking

- 381. Print all subarrays.
- 382. Count all subarrays.
- 383. Find sum of every subarray.
- 384. Find maximum subarray sum using brute force.
- 385. Find minimum subarray sum.
- 386. Find subarray with target sum.
- 387. Count subarrays with target sum.
- 388. Find longest subarray.
- 389. Find shortest subarray.
- 390. Find maximum average subarray.
- 391. Find subarray with maximum product.
- 392. Find subarray containing only positive numbers.
- 393. Find longest increasing subarray.
- 394. Find longest decreasing subarray.
- 395. Find maximum consecutive ones.
- 396. Find longest consecutive zeros.
- 397. Find longest alternating sequence.
- 398. Find equilibrium index.
- 399. Find prefix sum array.
- 400. Answer range sum queries.

STAGE 12: Strings From Baby Level

- 401. Print each character.
- 402. Count characters.
- 403. Reverse a string.
- 404. Check palindrome.
- 405. Count vowels.
- 406. Count consonants.
- 407. Count digits.
- 408. Count spaces.
- 409. Count uppercase.
- 410. Count lowercase.
- 411. Convert uppercase to lowercase.
- 412. Convert lowercase to uppercase.
- 413. Toggle case.

- 414. Remove spaces.
- 415. Replace spaces.
- 416. Count words.
- 417. Find longest word.
- 418. Find shortest word.
- 419. Reverse every word.
- 420. Reverse word order.

STAGE 13: String Logic

- 421. Count frequency of characters.
- 422. Find duplicate characters.
- 423. Find first repeating character.
- 424. Find first non-repeating character.
- 425. Find most frequent character.
- 426. Find least frequent character.
- 427. Remove duplicate characters.
- 428. Check anagram.
- 429. Check rotation.
- 430. Check if one string is substring of another.
- 431. Find all occurrences of a substring.
- 432. Count occurrences of a substring.
- 433. Find longest common prefix.
- 434. Find longest common substring.
- 435. Find longest palindromic substring.
- 436. Compress a string.
- 437. Decompress a string.
- 438. Check balanced characters.
- 439. Remove consecutive duplicates.
- 440. Find first unique character.

STAGE 14: Sorting From Scratch

- 441. Sort two numbers.
- 442. Sort three numbers.
- 443. Sort four numbers manually.
- 444. Understand swapping.
- 445. Bubble sort one pass.
- 446. Complete bubble sort.

- 447. Optimize bubble sort.
- 448. Selection sort.
- 449. Insertion sort.
- 450. Compare bubble and selection.
- 451. Sort ascending.
- 452. Sort descending.
- 453. Sort only even numbers.
- 454. Sort only odd numbers.
- 455. Sort positive and negative separately.
- 456. Find kth smallest.
- 457. Find kth largest.
- 458. Sort strings alphabetically.
- 459. Sort by string length.
- 460. Sort objects conceptually.

STAGE 15: Searching

- 461. Linear search.
- 462. Find first occurrence.
- 463. Find last occurrence.
- 464. Count occurrences.
- 465. Search in sorted array.
- 466. Binary search.
- 467. Find insertion position.
- 468. Find floor value.
- 469. Find ceil value.
- 470. Find square root using binary search.
- 471. Find first true condition.
- 472. Find last true condition.
- 473. Find peak element.
- 474. Search rotated sorted array.
- 475. Find minimum in rotated array.
- 476. Find duplicate using binary search ideas.
- 477. Find integer square root.
- 478. Find Nth root.
- 479. Search in matrix.
- 480. Search infinite sorted array conceptually.

STAGE 16: Mathematical Thinking for DSA

- 481. Count divisors.
- 482. Print divisors.
- 483. Sum divisors.
- 484. Check prime efficiently.
- 485. Sieve of Eratosthenes.
- 486. Prime factorization.
- 487. Count prime factors.
- 488. Check power of 2.
- 489. Check power of 3.
- 490. Check power of 4.
- 491. Fast exponentiation.
- 492. Modular arithmetic basics.
- 493. Calculate $(a + b) \bmod m$.
- 494. Calculate $(a * b) \bmod m$.
- 495. Find modular exponentiation.
- 496. Calculate GCD of array.
- 497. Calculate LCM of array.
- 498. Find trailing zeros in factorial.
- 499. Find number of digits in factorial.
- 500. Solve basic combinational counting.

How to Solve Every Question

1. Understand the problem and identify the input and expected output.
2. Solve a small example manually on paper.
3. Write the simplest brute-force approach first.
4. Implement the solution in Java.
5. Test normal cases and edge cases.
6. Improve the solution only after understanding the basic approach.
7. Learn and write the time complexity and space complexity.
8. Revisit difficult questions after 2–3 days.

Next DSA Stages

Questions 501–600: Recursion from absolute zero

601–700: Hashing and frequency problems

701–800: Two pointers

801–900: Sliding window

901–1050: Linked List

1051–1150: Stack

1151–1220: Queue and Deque

1221–1400: Trees

1401–1500: Heap and Priority Queue

1501–1700: Graphs

1701–1800: Backtracking

1801–2100+: Dynamic Programming